

PALLAB DEY

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SUMMARY

Computer Science graduate (CGPA 3.96/4.00) turning complex datasets into decisions companies act on, from uncovering an 11.3% margin decline masked by revenue growth, to building fraud-detection rules that flagged a \$518K high-risk transaction chain, to automating reporting pipelines that remove manual dashboard checks entirely. Practical experience across SQL, Power BI, Databricks, and Python.

SKILLS

Programming & Analysis: SQL (MS SQL Server), Query Optimization, Python (Pandas), ETL Pipelines, Data Modeling

BI & Visualization: Excel, Power BI, DAX, Power Query, Dashboard Design

Cloud & Automation: Databricks, Docker, n8n, Git/GitHub, Vector Databases (Pinecone)

PROJECTS

[Maven Toys Sales & Inventory Analysis](#) | SQL Server, Power BI, DAX, Power Query | [Live Dashboard](#)

- Built an end-to-end sales and inventory analysis on **829K transaction records** across **50 retail stores** by modeling a star-schema in SQL Server, uncovering that gross margin fell **11.3%** despite **30.9%** revenue growth.
- Designed a **5-page interactive Power BI dashboard** with drill-through store detail pages and custom DAX measures (YoY growth, margin %, stockout risk), giving stakeholders store-level and category-level visibility.
- Identified that the top revenue category (**Toys, 35%** of revenue) carried the lowest margin (**21%**) while Electronics led on margin, translating the finding into a category-mix and inventory recommendation.

[Automated Transaction Monitoring Pipeline](#) | Databricks (PySpark/SQL), n8n, Telegram Bot API

- Engineered a cloud data pipeline in Databricks to process **6.3M synthetic mobile money transactions**, building a star-schema model to support daily and weekly KPI reporting.
- Developed **4 explainable, rule-based risk triggers** (fraud rate breach, transfer-to-cash-out chains, volume collapse, transaction spikes) validated against ground-truth fraud labels, surfacing a high-risk laundering chain worth **~\$518K**.
- Automated reporting via **3 cron-scheduled n8n workflows**: a daily KPI monitor, a weekly performance summary and a conditional alert engine that stays silent unless a risk trigger fires, delivering all reports directly to Telegram without manual dashboard checks.

[RAG Pipeline & AI Chatbot](#) | n8n, Google Gemini Embeddings, Pinecone, Google Gemini

- Designed a Retrieval-Augmented Generation chatbot workflow in n8n, connecting Google Drive as a live document source with an automated ingestion pipeline covering download, recursive character text splitting, and embedding.
- Configured an AI Agent node with a Google Gemini chat model, conversational memory, and a Pinecone retrieval tool, allowing the chatbot to answer natural language questions by querying indexed documents in real time.

EDUCATION

Metropolitan University, Sylhet | 2021 - 2025

B.Sc. in Computer Science and Engineering — CGPA: 3.96 / 4.00

CERTIFICATIONS

Data Analyst Associate — DataCamp

Data Analyst in Databricks — DataCamp

ACHIEVEMENTS

- Recipient of the Chairman's Scholarship and Vice Chancellor's Scholarship at Metropolitan University.
- Solved **100+** SQL problems across StrataScratch, DataLemur, HackerRank, and LeetCode — documented in the SQL Practice Portfolio on [GitHub](#).